

Homework 4- Solutions

Problem 1. Find the volume under the surface $z = \frac{\ln(x) \ln(y)}{4xy}$ and above the rectangular region $1 \leq x \leq e^2, 1 \leq y \leq e^2$.

Solution.

$$\begin{aligned} \int_1^{e^2} \int_1^{e^2} \frac{\ln(x) \ln(y)}{xy} dx dy &= \frac{1}{4} \int_1^{e^2} \frac{\ln(x)}{x} dx \int_1^{e^2} \frac{\ln(y)}{y} dy \\ &= \frac{1}{4} \left. \frac{(\ln(x))^2}{2} \right|_1^{e^2} \left. \frac{(\ln(y))^2}{2} \right|_1^{e^2} \\ &= \frac{1}{16} ((\ln(e^2))^2 - 0)((\ln(e^2))^2 - 0) = 1 \end{aligned}$$

Problem 2. Suppose a discrete random variable K has the following pmf:

$$p_K(k) = \frac{1}{2^k} \quad ; k = 1, 2, 3 \dots$$

- (a) Find cumulative distribution function $F_K(k)$.
- (b) Find $E[2K - 1]$.

We know if our discrete random variable takes only 6 values, then our cumulative distribution will have 7 pieces (one piece for what happens before the random variable takes the first nonzero value and then 6 jumps, for each value that random variable takes). Think about for example when we roll a fair dice once. Then your cdf will look like

$$F_X(x) = \begin{cases} 0 & x < 1 \\ F(1) & 1 \leq x < 2 \\ F(2) & 2 \leq x < 3 \\ F(3) & 3 \leq x < 4 \\ F(4) & 4 \leq x < 5 \\ F(5) & 5 \leq x < 6 \\ F(6) & 6 \leq x \end{cases}$$

(write down the whole thing for one dice roll and see if the above is true?) So if we know the CDF at the discrete values then we can write down the cdf function. For example the above example tells that for any value n , if we want to find $F(x)$ where x is a point in interval $[n, n+1)$ then $F(x) = F(n)$.

For this example your random variable can take an infinite set of values, so your cdf will have infinite pieces, but from the above example we learn that

$$F_X(x) = \begin{cases} 0 & x < 1 \\ F(n) & n \leq x < n+1; n = 1, 2, 3, \dots \end{cases} \quad (1)$$

Now we should try to find $F(n)$.

$$\begin{aligned} F(n) &= P(X \leq n) = 1 - P(X > n) = 1 - \sum_{i=n+1}^{\infty} \left(\frac{1}{2}\right)^i \\ &= 1 - \frac{\left(\frac{1}{2}\right)^{n+1}}{1 - \frac{1}{2}} = 1 - \frac{\left(\frac{1}{2}\right)^{n+1}}{\frac{1}{2}} = 1 - \left(\frac{1}{2}\right)^n \end{aligned}$$

Hence, your cdf will have the following form

$$F_X(x) = \begin{cases} 0 & x < 1 \\ 1 - \left(\frac{1}{2}\right)^n & n \leq x < n+1; n = 1, 2, 3, \dots \end{cases} \quad (2)$$

Part (b). $E[2K - 1] = 2E[K] - 1$

$$\begin{aligned} E[K] &= \sum_{k=1}^{\infty} k \cdot P(K = k) = \sum_{k=1}^{\infty} k \cdot \frac{1}{2^k} \\ &= \frac{1}{2} \sum_{k=1}^{\infty} k \cdot \left(\frac{1}{2}\right)^{k-1} \end{aligned}$$

If we have $G(q) = \sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$. Then $G'(q) = \sum_{k=1}^{\infty} kq^{k-1} = \frac{1}{(1-q)^2}$. Now if we plug $q = \frac{1}{2}$, we have

$$E[K] = \frac{1}{2} \sum_{k=1}^{\infty} k \cdot \left(\frac{1}{2}\right)^{k-1} = \frac{1}{2} \cdot \frac{1}{\left(1 - \frac{1}{2}\right)^2} = \frac{4}{2} = 2$$

Therefore, $E[2K - 1] = 2E[K] - 1 = 4 - 1 = 3$

Problem 3. Assume pdf of a random variable X is given as below:

$$f(x) = \begin{cases} \frac{1}{(b-a) + (d-c)} & x \in [a, b] \cup [c, d] \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

(a) Find cumulative distribution function $F_X(x)$.

(b) Find $\text{Var}(X)$.

$$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{(b-a) + (d-c)}, & a \leq x < b \\ \frac{b-a}{(b-a) + (d-c)} & b \leq x < c \\ \frac{b-a + x-c}{(b-a) + (d-c)} & c \leq x < d \\ 1 & x \geq d \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} xf(x)dx = \frac{1}{(b-a) + (d-c)} \left(\int_a^b xdx + \int_c^d xdx \right) \quad (4)$$

$$= \frac{1}{(b-a) + (d-c)} \left(\left(\frac{b^2}{2} - \frac{a^2}{2} \right) + \left(\frac{d^2}{2} - \frac{c^2}{2} \right) \right) \quad (5)$$

$$= \frac{b^2 - a^2 + d^2 - c^2}{2(b-a + d-c)} \quad (6)$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x)dx = \frac{1}{(b-a) + (d-c)} \left(\int_a^b x^2 dx + \int_c^d x^2 dx \right) \quad (7)$$

$$= \frac{1}{(b-a) + (d-c)} \left(\left(\frac{b^3}{3} - \frac{a^3}{3} \right) + \left(\frac{d^3}{3} - \frac{c^3}{3} \right) \right) \quad (8)$$

$$= \frac{b^3 - a^3 + d^3 - c^3}{3(b-a + d-c)} \quad (9)$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 \quad (10)$$

$$= \frac{b^3 - a^3 + d^3 - c^3}{3(b - a + d - c)} - \frac{(b^2 - a^2 + d^2 - c^2)^2}{4(b - a + d - c)^2} \quad (11)$$

$$= \frac{4(b - a + d - c)(b^3 - a^3 + d^3 - c^3) - 3(b^2 - a^2 + d^2 - c^2)^2}{12(b - a + d - c)^2} \quad (12)$$

Problem 4. There are two coins, one with probability p_1 of Heads and the other with probability p_2 of Heads. One of the coins is randomly chosen (with equal probabilities for the two coins). It is then flipped $n \geq 2$ times. Let X be the number of times it lands Heads.

- (1) What is the PMF of X ? (Hint: Find what values X can take and for each $X = x$ find $P(X = x)$)
- (2) What is distribution of X when $p_1 = p_2$.

C_1 =Coin 1, $P(H|C_1) = p_1$
 C_2 =Coin 2, $P(H|C_2) = p_2$

(1) X = number of heads in n coin flips.
 X can take values $0, 1, 2, \dots, n$

$$\begin{aligned} p(X = k) &= p(X = k|C_1) p(C_1) + p(X = k|C_2) p(C_2) \\ &= \binom{n}{k} p_1^k (1 - p_1)^{n-k} \frac{1}{2} + \binom{n}{k} p_2^k (1 - p_2)^{n-k} \frac{1}{2} \\ &= \frac{\binom{n}{k}}{2} (p_1^k (1 - p_1)^{n-k} + p_2^k (1 - p_2)^{n-k}) \end{aligned}$$

(2) If $p_1 = p_2$, then

$$p(X = k) = \frac{\binom{n}{k}}{2} (2p_1^k (1 - p_1)^{n-k}) = \binom{n}{k} p_1^k (1 - p_1)^{n-k}$$

$X \sim \text{Bin}(n, p_1)$ where $p_1 = p_2$.

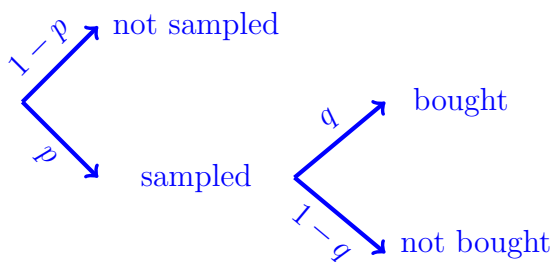
Just for your info, if $p_1 \neq p_2$, let

$$I = \begin{cases} 1 & \text{if coin 1 is chosen} \\ 0 & \text{if coin 2 is chosen} \end{cases}$$

Now $X = X_1 I + (1 - I)X_2$ where $X_1 \sim \text{Bin}(n, p_1)$ and $X_2 \sim \text{Bin}(n, p_2)$. It cannot be simplified further, but if you need to calculate expected value of X you can use this notation.

Problem 5. Sarah goes for shopping cookies to a bakery that sells n different types of cookies. Sarah samples a cookie with probability p (independently for each type). If a cookie type is sampled, then she buys that type with probability q , independently of other cookie types. Let Y be the number of cookie types that are sampled, and let X be the number of cookie types that Sarah buys. Find the distribution of Y and the distribution of X . Explain your answer.

Solution. For each cookie type we have the following tree diagram.



$P(\text{Sampling a cookie type}) = p$, and $P(\text{buying a cookie type}) = pq$

$Y = \text{number of cookie type sampled}$

Samplings are independent, in total we have n types, and probability of sampling remains same, hence it is a binomial. That is $Y \sim \text{Bin}(n, p)$.

$X = \text{number of cookie types bought}$

For each cookie type Sarah independently makes a choice, whether to buy it or not, probability of buying is same for all the cookie type, in total there are n types. Hence, $X \sim \text{Bin}(n, pq)$

Problem 6. Sarah immigrated to a new city, and would like to make friends with her new neighbours.

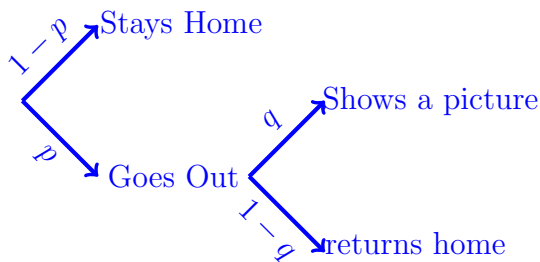
On any particular day i , she feels shy with probability $1 - p$ ($0 < p < 1$) and stays home; or, with probability p , she goes out and visits the i -th house in her neighborhood. At any house that she visits, either:

- (i) someone is at the house and answers the door; this happens with probability q (where $0 < q < 1$). In that case, Sarah shows them a picture of her hometown;
- (ii) no one is at the house and Sarah returns home.

We assume that the collection of all events of the form Sarah stays home on day i and Someone is at the i -th house on day i , for $i = 1, 2, \dots$, are (mutually) independent.

- (6-1) What is the probability that Sarah gives out a picture on the first day?
- (6-2) What is the probability that she gave out exactly k pictures on the first n days? where k and n are some fixed integers with $n \geq 1$ and $0 \leq k \leq n$.
- (6-3) What is the probability that she stayed home on day 1, given that she did not give out a picture on that day?
- (6-4) Within the first week (7 days), what is the probability that there were no days in which Sarah went out but did not give out a picture of her hometown?

Solution. Generally in day i we have



- (6-1) Using multiplication rule from the above diagram $p \cdot q$. This is the probability of giving a picture out in a given day.
- (6-2) X = Number of pictures that she gives out in n day, then $X \sim \text{Bin}(n, p \cdot q)$. Hence,

$$P(X = k) = \binom{n}{k} (p \cdot q)^k (1 - p \cdot q)^{n-k}$$

- (6-3) $P(\text{Staying home} | \text{no picture is given that day}) = \frac{1 - p}{1 - P(\text{giving out a picture on that day})} = \frac{1 - p}{1 - pq}$
- (6-4) No days in which Sarah goes out, but does not give a picture of her home town. In a specific day, there are two ways that this happens; either Sarah stays home ($1 - p$), or she goes out and gives a picture of her home town $p \cdot q$. Hence, for each day this happens with probability $(1 - p + pq)$. Now we have 7 days, and what happens in each day is independent from another day, hence, the desired scenario happens with probability $(1 - p + pq)^7$.